MER1 - Usuarios, estudiantes y rutas

Tablas base de identidad escolar, contactos, rutas y paradas

Usuarios y estudiantes

users

PK id : UUID

email : VARCHAR(100) UNIQUE

password_hash : VARCHAR(255)

is_active : BOOLEAN

email_verified : BOOLEAN

two_factor_enabled : BOOLEAN

created_at · updated_at

1 0..* responsable

students

PK id : UUID

FK user_id -> users.id

full_name : VARCHAR(100)

school_grade : ENUM

birth_date : DATE

photo_data : TEXT

is_active : BOOLEAN

1

acudientes

0..*

autoriza

student_guardians

PK id : UUID

FK student_id -> students.id

FK user_id -> users.id

relationship_role : VARCHAR(30)

status : VARCHAR(20)

UNIQUE (student_id, user_id)

emergency_contacts

PK id : UUID

FK student_id -> students.id

full_name · phone

relationship · is_primary

is_active : BOOLEAN

Rutas escolares

routes

PK id : UUID

route_name · description

origin_latitude · origin_longitude

destination_latitude · destination_longitude

FK created_by / updated_by

1

asignación

1

asignación

student_routes

PK/FK student_id -> students.id

PK/FK route_id -> routes.id

is_active : BOOLEAN

assigned_at : TIMESTAMPTZ

contiene

0..*

1..*

stops

PK id : UUID

FK route_id -> routes.id

stop_order · stop_name

latitude · longitude

PK = clave primaria · FK = clave foránea. created_by / updated_by referencian users.id cuando aparecen.

MER2 - Seguridad y control de acceso

Roles, permisos, sesiones y configuraciones de seguridad

Control de acceso basado en roles

users (referencia)

PK id : UUID

email · is_active

two_factor_enabled

roles

PK id : UUID

role_name : ADMIN | PARENT

description

0..1 — 0..1

0..1 — 1

11

asigna

inicia

0..* — 0..*

participa

actualiza

user_roles

PK/FK user_id -> users.id

PK/FK role_id -> roles.id

assigned_at : TIMESTAMPTZ

user_sessions

PK id : UUID

FK user_id -> users.id

token · device_identifier

platform · source_ip

session_started_at · session_ended_at

session_status : ENUM

device_confirmation_token

device_confirmation_return_url

permissions

PK id : UUID

permission_name UNIQUE

description

asigna

0..*

role_permissions

PK/FK role_id -> roles.id

PK/FK permission_id -> permissions.id

Políticas y configuración

password_policies

PK id : UUID

min_length · max_length

requires_uppercase/numbers/symbols

expiration_days

FK updated_by -> users.id

0..*

security_settings

PK id : UUID

setting_name UNIQUE

setting_value · description

FK updated_by -> users.id

0..*

user_sessions.session_status usa ACTIVE, EXPIRED o CLOSED. Contraseñas y tokens no se representan en claro en la aplicación.

MER3 - Seguimiento, zonas y notificaciones

Rastreo de trayectos, coordenadas GPS, zonas seguras y avisos a usuarios

Seguimiento GPS

users (referencia)

PK id : UUID

email · is_active

students (referencia)

PK id : UUID

FK user_id

full_name

routes (referencia)

PK id : UUID

route_name

users (referencia) 1 — 1 students (referencia) 0..* responsable

students (referencia) 1 — 1 routes (referencia)

students (referencia) 1 — 0..* safe_zones

students (referencia) 1 — 0..* trips

routes (referencia) 1 — 0..* trips

safe_zones 0..1 — 0..* trips

trips 1 — 1..* coordinates

safe_zones

PK id : UUID

FK student_id -> students.id

zone_name · zone_type · color

address

latitude · longitude · radius_meters

inactivity_alert_seconds · is_active

trips

PK id : UUID

FK student_id -> students.id

FK route_id -> routes.id

trip_started_at · trip_ended_at

status : trip_status_enum

had_route_deviation

coordinates

PK id : UUID

FK trip_id -> trips.id

latitude · longitude

recorded_at · speed_kmh

stopped_seconds

Notificaciones

notification_settings

PK id : UUID

FK user_id -> users.id

alerts: delay/route/arrival/inactivity

channels: push/email/SMS

users (referencia) 1 — 0..1 notification_settings

notifications

PK id : UUID

FK trip_id -> trips.id (nullable)

FK safe_zone_id -> safe_zones.id (nullable)

event_type · message

event_at · was_sent

trips 0..1 — 0..* notifications

users (referencia) 1 — 0..* notifications

notification_receipts

PK id : UUID

FK notification_id -> notifications.id

FK user_id -> users.id

was_read · received_at · read_at

notifications 1 — 0..* notification_receipts

users (referencia) 1 — 0..* notification_receipts

notificaciones puede originarse en un trayecto o una zona segura; los recibos son individuales por usuario autorizado.

MER4 - Dispositivos y registros operativos

Tokens de notificación, dispositivos vinculados y trazabilidad de actividad

Dispositivos vinculados

users (referencia)

PK id : UUID

email · is_active

students (referencia)

PK id : UUID

full_name · is_active

0..1

0..1

1

0..1

1

registra

vincula có

audita

0..*

0..*

user_devices

PK id : UUID

FK user_id -> users.id

expo_push_token UNIQUE

platform · device_name

is_active · last_used_at

student_devices

PK id : UUID

FK student_id -> students.id

device_identifier UNIQUE

platform · device_name

is_active · linked_at · last_used_at

Auditoría y registros

activity_logs

PK id : UUID

FK user_id -> users.id (nullable)

action_type · description

source_ip · metadata · created_at

0..*

audit_logs

PK id : UUID

FK user_id -> users.id (nullable)

action · description · application

source_ip · metadata · created_at

0..*

error_logs

PK id : UUID

FK user_id -> users.id (nullable)

error_type · description · stack_trace

source_ip · metadata · created_at

0..*

student_devices almacena el identificador asociado al código/QR de vinculación. Los registros admiten user_id nulo para eventos técnicos.